Nayeon Kim

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Research Interest

Improving fine-grained perception and spatial reasoning in multimodal models

- Agent, Autonomous Driving, Embodied AI

Education

University of Minnesota - Twin Cities *Sep 2026 – Present*
Ph.D. in Computer Science

- Advisor: Prof. Yao-Yi Chiang

Kookmin University (KMU) *Mar 2015 – Feb 2019*
B.S. in Computer Science, Automotive Software Design

- GPA: 4.19/4.5 (**Rank : 1/49**)

Work Experience

Samsung Electronics AI Center, Korea - *AI Researcher* *Mar 2019 – Aug 2026*

- Developed a generative model for industrial process automation by predicting time-series data.
- Developed computer vision models for high-level semantic and geometric understanding of road scenes : Online HD Map construction (NeurIPS '24), 3D lane detection (ICASSP '23)

Korea Electronics Technology Institute (KETI), Korea - *Research Intern* *Dec 2017 – Oct 2018*

- Developed a computer vision model for monocular 3D perception: Monocular depth prediction (KSAE '18)

Publications

Spatial Query Execution for Multimodal Spatial Reasoning *Under Review*
Jiyoon Pyo, Dongwon Jung, Zekun Li, Nayeon Kim, Muhao Chen, Yao-Yi Chiang

Unveiling the Hidden: Online Vectorized HD Map Construction with Clip-Level Token Interaction and Propagation *NeurIPS 2024*
Nayeon Kim*, Hongje Seong*, Daehyun Ji, Sujin Jang

D3DLD: Depth-aware Voxel Space Mapping for Monocular 3D Lane Detection with Uncertainty *ICASSP 2023*
Nayeon Kim*, Moonsub Byeon*, Daehyun Ji, Dokwan Oh

Unsupervised Depth Prediction and Camera Motion Estimation in a Dynamic Environment *KSAE 2018*
Nayeon Kim, MinGyu Park, Youngbae Hwang

Patents

Method for Automated Analyzing Semiconductor Samples Using Future Frame Generation *In Progress*
Nayeon Kim, Jiwon Lee, Zyup Jeong, Sujin Jang

Method for Automated Preparation of Semiconductor Analysis Samples Using a Self-Learning AI Decision Model *In Progress*
Jiwon Lee, Nayeon Kim, Zyup Jeong, Sujin Jang

Method and Apparatus with Vehicle Driving Control *US-Application No. 18/936,286*
Hongje Seong, Nayeon Kim, Sujin Jang, Daehyun Ji filed on 11/04/2024

Method and Apparatus with Vector Map Learning and Generation *US-Application No. 18/605,119*
Nayeon Kim, Sujin Jang, Dae Ung JO filed on 03/14/2024

Method and Apparatus with Lane Generation *US-Application No. 17/862,821*
Nayeon Kim, Moonsub Byeon, Dokwan Oh, Daehyun Ji filed on 07/12/2022

Method and Apparatus for Determining Slope of Road Using Side View Camera of Vehicle *US-Application No. 17/685,917* filed on 03/03/2022
Moonsub Byeon, Nayeon Kim, Daehyun Ji

Awards & Honors

Employee of the Year *Samsung Electronics 2025*
CEO Commendation Award ('23.4Q DS Vision Awards) *Samsung Electronics 2024*
Business Division Commendation Award (1H 2022) *Samsung Electronics 2022*

Projects

Automated Semiconductor Cross-Section Analysis *Samsung Electronics Jan 2025 – Aug 2026*

- Developed an automated system using generative models to analyze semiconductor cross-sectional images and automatically determine optimal stopping points during the milling process.

Large Language Model Alignment Tuning *Samsung Electronics June 2024 – Dec 2024*

- Conducted research on alignment tuning methods to enhance LLM performance and safety for internal model development.

Online Vectorized HD Map Construction *Samsung Electronics Jan 2023 – May 2024*

- Developed an algorithm of vectorized HD map construction that can predict road information in challenging driving road scenes.
- Developed an algorithm that effectively constructs vectorized HD map using only a camera, by applying knowledge distillation from the multi-modal fusion of lidar and camera.

Lane Detection *Samsung Electronics Sep 2019 – Dec 2022*

- Developed an algorithm to predict 3-dimension lane information from a single image that is robust to various driving environments.
- Developed a 2-dimension lane fitting algorithm that can be implemented in real vehicles.

Unsupervised Depth Prediction *KETI Dec 2017 – Oct 2018*

- Developed an algorithm to improve the performance of the depth map by predicting the movement between images through the flow network.

Academic Service

Conference Reviewer: NeurIPS *2025 - 2026*

Skills

Programming Languages: Python, C++, C
Frameworks: PyTorch, ROS, OpenCV
Tools: Linux, Git, Docker
Languages: Korean(Native), English